

MATEO WU CHEN

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EDUCATION

Northeastern University, College of Engineering – Boston, MA **Dec 2026**
Candidate for Bachelor of Science in Computer Engineering, Minor in Computer Science
Awards and Activities: Dean's list, NU Robotics Club, SASE, Forge GPA: 3.5

TECHNICAL SKILLS

Electronics: Arduino, Digital Multimeter, FPGAs, Laser Cutting, Oscilloscope, PCB Design, Raspberry Pi, Soldering, 3D Printing
Software: Altium Design, C++, C#, Git, Java, KiCAD, Linux, LTspice, MATLAB, Python, System Verilog
Languages: Native Spanish, Fluent English, Basic Mandarin

EXPERIENCE

Tatum Robotics – Watertown, MA **January 2025 – July 2025**
Embedded Systems & Hardware Engineer Coop

- Designed and tested PCBs in Altium Designer for the TatumT1 Assistive Touch Hand, including power and button boards for component integration and user interaction, and a custom RGB illumination system that significantly improved feedback functionality, contributing to a 100% functional pass rate in final prototypes
- Migrated legacy PWM control from RPi.GPIO to gpiod on Raspberry Pi 5, programmed and debugged Python and C# applications, integrated I2C devices, and optimized software-hardware interaction to improve system response time by 80%
- Supported robot assembly and ongoing development, executing comprehensive electrical testing protocols on prototype units using a digital multimeter for voltage measurements and an oscilloscope for signal analysis, achieving a 95% fault detection rate and ensuring compliance with industry performance and safety standards

Northeastern University SINRG – Boston, MA **January 2025 – Present**
Undergraduate Research Assistant - Spatial Intelligence Research Group

- Contributed to the development of a prototype AI glasses system by designing custom 4-layer PCBs in Altium Designer, assisting in schematic capture and circuit design for power regulation and ESP-32 microcontroller interfaces, and applying EMI reduction techniques to improve signal integrity and system reliability
- Evaluated hardware requirements and ensure compatibility with embedded systems and optics modules, focusing on efficient component layout for wearable integration and sensor integration to support real-time AI functionality

Northeastern University NUCAR – Boston, MA **May 2023 – December 2024**
Undergraduate Research Assistant – Computer Architecture Research Group

- Optimized data processing and algorithm performance by implementing parallel computing with Pthreads in C++, enhancing data synchronization for multi-threaded applications, and conducting performance analyses using varied thread counts and timing mechanisms, resulting in a ~80% reduction in runtime for large-scale datasets
- Leveraged the Discovery Cluster for high-performance computing, creating and executing Slurm job scripts to allocate GPU resources and benchmark single and double precision floating-point calculations across GPUs and CPUs
- Implemented GPU-accelerated matrix multiplication in CUDA C++, achieving up to +1000x performance improvement compared to CPU-based methods, and optimized multi-threaded CPU computations using OpenMP

PROJECTS

Custom Hot-swappable Macro Pad Keyboard PCB **June 2025 – July 2025**
Created and assembled custom 4 layers mechanical macro pad PCB using Altium Designer, from schematic to PCB Layout & routing, applying design techniques by integrating features like fill zones and EMI reduction

Pill-Pal-4-Life Assistive Robot, Forge **September 2024 – December 2024**
Collaborated on the development of a compact, user-friendly medication reminder robot for children, featuring a multifunctional touchscreen interface, designed to improve medication adherence through engaging design and ease of use.
Developed authentication system by programming a fingerprint sensor using C++ and Arduino Mega to ensure secure, scheduled pill dispensing accessible for child, contributing on system electronics and LED interaction design

Pokémon Combat Game, *Computing Fundamentals for Engineers* **October 2023 – December 2023**
Built a Pokémon clone game in Python using OOP, PokeAPI, sound effects, and UI design like background music, GUI and sound effects to enable user battles and enhance gameplay, ensuring an immersive and user-friendly gaming experience

INTEREST

Guitar, Gundam Plastic Models (GUNPLA), PC Building, Physical Fitness, Mechanical Keyboard Enthusiast, Reading, Cooking